

HerWay: A Data-Driven System for Chicago Neighborhood Awareness

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Abstract—Urban safety is shaped not only by recorded incidents but also by how people experience and perceive public spaces. Individuals navigating unfamiliar urban environments often lack access to contextual information that extends beyond official statistics, making it difficult to understand how neighborhoods are experienced by residents. We present HerWay, an interactive urban awareness platform designed to support neighborhood exploration and contextual understanding across Chicago. HerWay integrates community discussion, public safety records, and municipal service data into neighborhood-level profiles that can be explored through an interactive map and conversational chatbot.

The platform combines Reddit community discussions, Chicago crime records, and 311 service requests to capture both official conditions and lived experiences. These data sources are processed and aggregated into profiles covering all 77 Chicago community areas. Users can explore neighborhoods through a Leaflet-based mapping interface, search functionality, neighborhood information panels, and a GPT-4o-powered chatbot that supports comparison, exploration, and perception-gap reasoning.

Key findings demonstrate that community perception and official records do not always align. Female-perspective Reddit posts expressed higher levels of fear than overall community discussions, while infrastructure disparities identified through 311 data revealed longer service delays in historically underserved neighborhoods. HerWay surfaces these patterns through a source-agreement framework that highlights where community concerns and official indicators converge or diverge.

By combining multiple urban data sources within a single platform, HerWay demonstrates how data-driven systems can improve access to contextual neighborhood information and support more informed exploration of urban environments.

<https://her-way-soremo.vercel.app>.

Index Terms—urban safety, sentiment analysis, NLP, named entity recognition, women’s safety, Chicago, multi-source integration, equity, chatbot

EXECUTIVE SUMMARY

What is this project about?

Have you ever changed your route home—not because something happened, but because of how a street *felt*? That question is at the heart of HerWay.

HerWay is a platform that helps people understand what Chicago neighborhoods are actually like to move through—not just what the crime statistics say, but how residents *experience* them. It brings together three kinds of information: what people post about their neighborhoods on Reddit (community voice), official crime data from the Chicago Police Department, and city service complaints (311 data) about things like broken street lights and abandoned buildings. These three sources are combined into a safety profile for each of Chicago’s 77 neighborhoods, accessible through an interactive map and a conversational chatbot that answers questions in plain language.

Why does it matter?

A student who has lived in Chicago for ten years knows instinctively which streets feel fine after dark. A student who arrived six months ago from another country has no such accumulated knowledge. Women who experience street harassment rarely report it to the police—making that experience invisible in official crime data. This is an *equity problem*: safety intelligence is unevenly distributed, and the people who need it most often have the least of it. HerWay is an attempt to rebalance that.

Who are the stakeholders?

Primary users: Women, international students, and newcomers navigating Chicago independently.

Secondary beneficiaries: City planners, community organisations, and researchers who want to understand where official data and community experience diverge.

Critical participants: The Reddit community members whose posts form HerWay’s community voice layer; the City of Chicago, whose open data portals make crime and 311 data available; and the Illinois Institute of Technology SoReMo Research Fellowship, which supported this work.

Methods, in plain language

HerWay uses four main techniques. The first is *senti-ment analysis*—a way of teaching a computer to read the emotional tone of text, so it can tell whether a Reddit post is expressing fear, seeking information, or reassuring others. The second is *named entity recognition*—a way of identifying place names in text, which we adapted specifically for Chicago’s neighbourhood names because standard tools did not recognise them. The third is a *weighted scoring formula* that combines all three data sources into a single, comparable safety profile per neighbourhood. The fourth element is the platform itself. Rather than presenting these results as static charts, HerWay delivers them through a conversational assistant that reasons about the gap between what the crime data records and what residents express. Its guiding logic is comparison: when community concern runs ahead of recorded incidents, or recorded incidents run ahead of community concern, that divergence is treated as the most informative thing to surface, in plain language and without ever labelling a neighbourhood “safe” or “unsafe.”

Key findings

Four findings stand out.

First, women express safety-related fear on Reddit at significantly higher rates than men: 28.5% of female-perspective posts vs. 23.5% for male-perspective posts. At night, the female fear rate reaches 33.3% vs. 19.5% for men—a 13.8 percentage-point gap that is entirely invisible in official crime records.

Second, Chicago’s infrastructure maintenance is unequal. Vacant building complaints take 41 days on average to resolve, compared to 7 days for street light outages. The neighbourhoods waiting longest are predominantly on Chicago’s South and West sides—the same ones with the highest crime counts and Reddit fear signals.

Third, when community fear and crime data disagree, that disagreement is itself the most informative finding. A neighbourhood where people report feeling unsafe despite moderate crime data may be experiencing harassment that never gets reported. This *source agreement signal* is HerWay’s primary novel contribution. A fourth result concerns the platform rather than the data. When asked

a deliberately loaded question such as “How bad is Englewood really?”, the assistant answered in neutral, data-grounded terms instead of repeating the stigmatising frame, demonstrating that the perception-gap reasoning and language constraints at the core of the design hold up in practice. This is what distinguishes HerWay from a crime map: it does not just show numbers, it interprets the distance between numbers and lived experience.

What equity issue was addressed?

HerWay directly addresses the inequitable distribution of safety information, making community-grounded safety knowledge accessible to those who would otherwise lack it. It also surfaces a concrete infrastructure equity finding: lower-income communities on Chicago’s South and West sides wait measurably longer for city services, a disparity that three independent data sources simultaneously confirm.

What equity issue was not addressed, and why?

Reddit users are not representative of all Chicago residents. They skew younger, more English-speaking, and more tech-savvy. The voices of elderly residents, non-English-speaking communities, and lower-income residents—the groups who may need HerWay most—are underrepresented in the data that informs it. A neighbourhood like Chinatown or a predominantly Spanish-speaking area may have rich safety discourse happening in spaces our pipeline did not capture. This is a genuine limitation, not a technical failure: addressing it would require multilingual data collection and community partnerships that were outside the scope of this semester-long fellowship project.

I. INTRODUCTION

A. Motivation and Personal Context

The idea for HerWay emerged from a shared personal experience. As international students at Illinois Institute of Technology, the authors navigated Chicago’s neighbourhoods with persistent uncertainty. In early months, moving in groups provided safety in numbers. As independent exploration became desired, a fundamental information gap became clear: no existing tool combined official safety statistics, city infrastructure data, and the community-grounded context that long-time residents develop through lived experience.

Google Maps tells you *how* to get somewhere—not how it feels to walk there at 9 pm as a young woman alone, that street lights are broken, or that residents are posting about recent harassment. This gap disproportionately affects

women, newcomers, and international students: an *equity gap in information access*.

B. Project Timeline

HerWay began in August 2025 as an independent exploration among the three authors in response to their shared experience navigating Chicago as newcomers. In January 2026, the project was selected for the Spring 2026 SoReMo Research Fellowship at IIT.

From January through March 2026, the team divided work into three parallel streams: Reddit data pipeline and NLP analysis (Sanjana Waghay), crime data analysis (Ananya Bhoolam Praveen), and 311 analysis, frontend, and system integration (Sharanya Mishra). A key analytical pivot occurred in February 2026 when the source agreement signal—where Reddit fear and crime data diverge—emerged as more valuable than the combined score alone, reshaping the system’s design philosophy.

By April 2026, all pipelines were merged, scoring was finalised, and the platform was deployed on Vercel. Report submission followed in May 2026.

C. Problem Statement

Existing tools present raw crime counts without contextual or perceptual nuance, or ignore safety altogether. Official crime data is structurally incomplete: street harassment, intimidation, and poorly lit streets rarely appear in police records [8]. Community discourse on Reddit captures exactly these informal signals, but no platform has integrated them with official data into a unified interface.

D. Contributions

- Reddit pipeline (1,218 posts) via public JSON endpoint
- Gazetteer-based NER for Chicago’s 80+ neighborhood names
- Transformer sentiment classification with 40-keyword safety lexicon
- Gender-aware analysis revealing 13.8 pp night-time fear gap
- TF-IDF neighborhood profiling for chatbot responses
- Weighted multi-source scoring with source agreement signal across all 77 community areas
- Deployed interactive map and GPT-4o chatbot at <https://her-way-soremo.vercel.app>

II. RELATED WORK

Urban safety analysis has traditionally relied on crime statistics and perception surveys—objective but incomplete, or rich but slow [8]. [1] demonstrated that Twitter-based fear reports captured perceptions invisible in police

data. [2] showed location-based social data correlates with crime patterns. [3] applied BERT classifiers to safety post detection at 87% accuracy. [9] showed urban imagery can predict neighbourhood socioeconomic profiles. [7] provided the theoretical foundation that women experience urban spaces differently from men—a premise our gender analysis tests empirically. HerWay extends this work by combining three independent sources, applying a women-centred lens, and delivering findings through a deployed conversational interface.

III. DATA SOURCES AND PREPROCESSING

HerWay integrates three datasets. Preprocessing standardised formats, parsed dates, mapped records to Chicago’s 77 community areas, and aggregated to neighbourhood level.

- **Reddit:** Informal safety narratives capturing harassment, lighting concerns, and perceived threat.
- **Crime (2025):** 200,000+ incidents from the Chicago Data Portal [10].
- **311 (2025):** Non-emergency resident complaints [11] capturing infrastructure quality and municipal responsiveness.

Data availability. The processed neighbourhood-level CSV files (Reddit fear ratios, crime profiles, 311 summaries) and analysis code are available in the project GitHub repository at <https://github.com/sanjana-3002/HerWay> and will be submitted for archival to the IIT Galvin Library alongside this report.

IV. 311 SERVICE REQUEST ANALYSIS

A. Motivation

A neighbourhood may have moderate crime yet feel unsafe because of broken street lights, abandoned buildings, or slow municipal response. Resolution time carries a specific equity signal: if lower-income communities wait far longer for repairs, that delay is a measurable form of urban inequality independent of crime.

B. Key Findings

Abandoned vehicle complaints dominate at 52.8%; street light outages account for 35.8% (Fig. 1)—precisely the conditions most cited as safety triggers in fearful Reddit posts, confirming a direct 311-to-perception signal.

Vacant building complaints take 41.1 days to resolve vs. 7.4 days for street light outages (Fig. 2)—a $5.6\times$ disparity. Neighbourhoods with the highest vacant building volumes are concentrated on Chicago’s South and West sides: the same areas leading in crime counts and Reddit fear

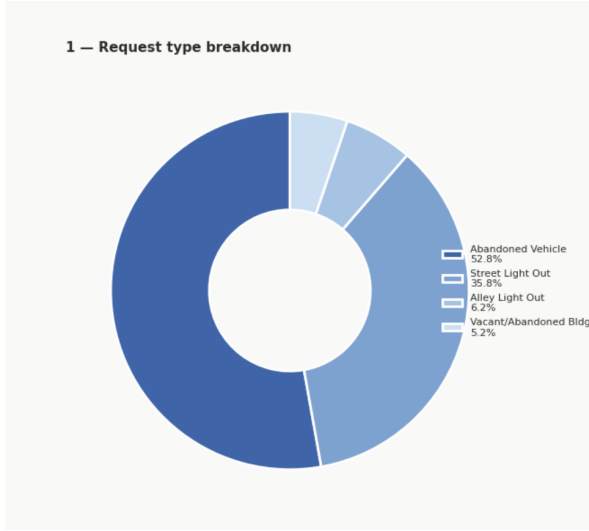


Fig. 1. 311 request type breakdown. Abandoned vehicles (52.8%) and street light outages (35.8%) dominate—the same conditions most cited in fearful Reddit posts.

ratios. This three-source convergence is the clearest equity finding in our analysis.

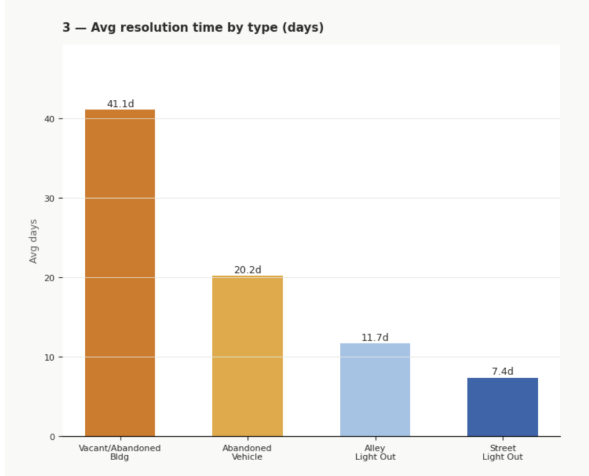


Fig. 2. Average 311 resolution time by request type. Vacant buildings take 5.6× longer than street light outages—a meaningful equity signal for underserved neighbourhoods.

V. CRIME DATA ANALYSIS

A. Data and Preprocessing

Crime data was filtered to 2025 [10]. Preprocessing derived: (1) temporal features; (2) violent/non-violent classification; (3) night crime classification (8 pm–5 am); (4) per-community aggregation into profiles with total incidents, crime type breakdown, violent and domestic percentages, arrest rate, and temporal peak attributes.

B. Key Findings

Theft leads citywide (Fig. 3), followed by battery and criminal damage. Violent crimes represent a smaller share of total volume but are geographically concentrated.

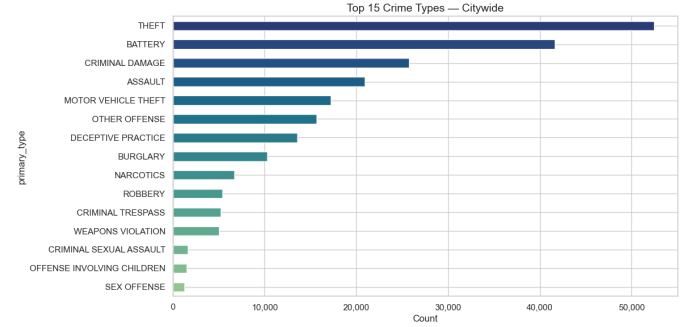


Fig. 3. Top 15 reported crime types citywide (2025). Theft, battery, and criminal damage dominate; sexual assault and weapons violations carry the highest severity weight in HerWay’s classification.

Austin leads at 11,056 incidents (Fig. 4). The Loop’s high count is theft-driven—a pattern raw counts alone cannot distinguish. Riverdale records the highest violent crime percentage (50.5%) with among the lowest total counts ($\approx 1,000$), illustrating the critical difference between volume and concentration.

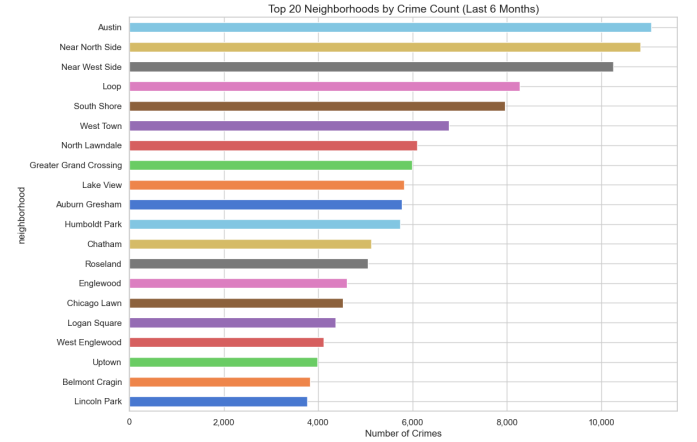


Fig. 4. Top 20 community areas by total incidents (2025). West and South side communities dominate. Austin leads at 11,056.

Crime follows a clear temporal pattern (Fig. 5): a midnight spike, a trough at 4–6 am, then a steady rise through the afternoon. This directly informed HerWay’s time-sensitive guidance.

C. Limitations

The dataset reflects *reported* crime only. Domestic violence, sexual assault, and crimes in communities with

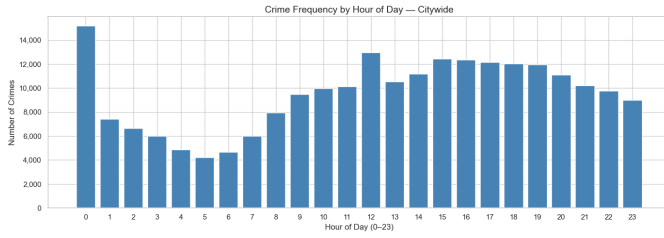


Fig. 5. Crime frequency by hour of day (2025). The midnight spike reflects late-night activity; the 4–6 am trough is the clearest lower-risk window across all community areas.

low institutional trust are systematically underreported [8]. Policing bias shapes the data: heavier police presence generates more records due to more encounters, not more crime. HerWay presents crime as one signal among several, always.

VI. REDDIT COMMUNITY DATA PIPELINE

A. Data Collection and Location Extraction

Reddit exposes structured data through a public JSON endpoint, requiring no authentication. We queried five subreddits (r/chicago, r/AskChicago, r/TwoXChromosomes, r/AskWomen, r/femalefashionadvice) with a two-second inter-request delay. **Total: 1,218 posts, 2010–2026.**

Standard NER models fail on hyper-local Chicago names—spaCy `en_core_web_sm` did not recognise “Pilsen” or “Back of the Yards”. We implemented **gazetteer-based NER**: a domain-specific dictionary of 80+ neighbourhood names matched via word-boundary regex, with a context filter for ambiguous toponyms. *Result: 499 of 1,218 posts (41%) tagged.* The pipeline is shown in Fig. 6.



Fig. 6. Reddit data collection pipeline, from 1,218 raw posts to 60 neighbourhood-level safety profiles.

B. Safety Lexicon and Sentiment Classification

A domain-specific lexicon of 40 keywords extracted a safety relevance score per post. General lexicons miss context-specific signals: “dark” and “alone” are neutral

TABLE I
DOMAIN-SPECIFIC SAFETY LEXICON (40 KEYWORDS TOTAL)

Category	Sample Keywords
Direct threat	mugged, attacked, stalked, grabbed
Environmental	dark, alone, sketchy, isolated
Harassment	catcall, followed, harassed, creepy
Emotional state	scared, afraid, terrified, uneasy
Violence	shooting, knife, assault, robbery

in standard dictionaries but unmistakably safety-relevant in urban navigation contexts (Table I).

Each post was classified into *Negative/Fear*, *Neutral/Concern*, or *Positive/Reassuring* using `cardiffnlp/twitter-roberta-base-sentiment`—a RoBERTa [5] transformer [6] fine-tuned on 58 million tweets [4]. The per-neighbourhood fear ratio:

$$r_n = \frac{|\{i : \hat{y}_i = \text{Fear}, n \in N_i\}|}{|\{i : n \in N_i\}|} \quad (1)$$

TF-IDF was applied per neighbourhood to surface distinctive vocabulary—transit references in Uptown, nightlife concerns in River North—feeding the chatbot’s contextually specific responses.

C. EDA Findings

The plurality of posts (44.7%) are Neutral/Concern (Fig. 7). Positive/Reassuring posts (31.1%) outnumber Negative/Fear posts (24.2%)—HerWay does not systematically inflate risk perception.

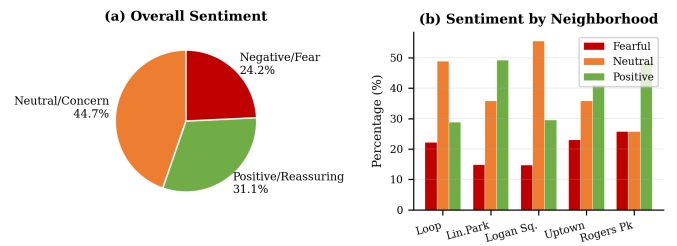


Fig. 7. Sentiment distribution across 499 geotagged posts (a) and by top 5 neighbourhoods (b). Positive posts outnumber fearful posts citywide.

The gender analysis surfaces the study’s most policy-relevant finding (Fig. 8). Female-perspective posts express fear at 28.5%—6.3 pp above the overall rate. At night, the female fear rate reaches **33.3%** vs. 19.5% for male-perspective posts—a **13.8 pp gap** consistent with [7] and entirely invisible in official crime records.

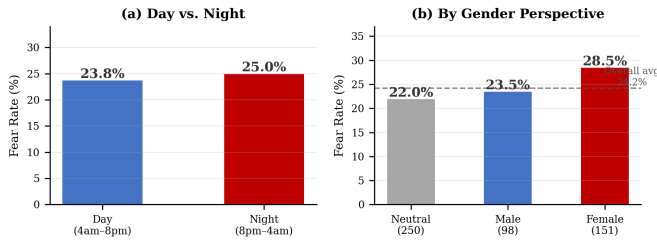


Fig. 8. Fear rate by time of day (a) and gender perspective (b). The 13.8 pp night-time gap between female (33.3%) and male (19.5%) perspectives is the project’s primary equity finding.

VII. MULTI-SOURCE INTEGRATION

A. Name Harmonization and Normalization

Reddit uses popular names that differ from official community area names (“Wicker Park” → “West Town”; “Pilsen” → “Lower West Side”). A mapping dictionary of 19 translations enables correct joins. Min-max normalization maps each source independently to $[0, 1]$:

$$\tilde{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

B. Weighted Scoring and Risk Classification

$$S_n = 0.40C_n + 0.25R_n + 0.20V_n + 0.10T_n + 0.05L_n \quad (3)$$

where C_n , R_n , V_n , T_n , L_n are normalized crime count, Reddit fear ratio, violent crime percentage, 311 complaints, and street lighting complaints. Table II details the rationale.

TABLE II
WEIGHTING SCHEME FOR COMBINED SAFETY SCORE

Signal	Wt.	Rationale
Crime count (C_n)	40%	Most objective, data-rich
Reddit fear (R_n)	25%	Lived community experience
Violent crime (V_n)	20%	Severity not just volume
311 complaints (T_n)	10%	Infrastructure proxy
Street lighting (L_n)	5%	Women’s safety factor

Neighbourhoods are classified High Risk ($S_n \geq 0.65$), Medium Risk ($0.40 \leq S_n < 0.65$), or Lower Risk ($S_n < 0.40$), yielding 3, 26, and 48 neighbourhoods. Table III shows the top 5.

TABLE III
TOP 5 NEIGHBOURHOODS BY COMBINED SAFETY SCORE

Neighbourhood	S_n	Reddit	Crime	Risk
Austin	0.772	34.8%	11,056	High
Gr. Grand Crossing	0.695	100%	5,992	High
Auburn Gresham	0.672	100%	5,778	High
Chicago Lawn	0.623	100%	4,522	Medium
South Shore	0.604	50.0%	7,963	Medium

C. Source Agreement Signal

The **source agreement signal** compares R_n with C_n to surface divergence between community perception and official data (Table IV). The third and fourth states are most analytically valuable: they reveal information neither source alone could provide. A neighbourhood with high community fear but moderate crime may be experiencing unreported harassment. One with elevated crime but calm community perception may be genuinely improving before statistics reflect it. This signal is surfaced in every map popup and anchors the chatbot’s reasoning.

TABLE IV
SOURCE AGREEMENT SIGNAL: FOUR STATES

R_n	C_n	Interpretation
High	High	Both signal concern. Strongest actionable finding. <i>E.g. Austin.</i>
Low	Low	Consistent reassurance. <i>E.g. Lincoln Park.</i>
High	Low	Community more concerned than data suggests. Possible unreported harassment.
Low	High	Data elevated but community calm. Possible genuine improvement.

VIII. HERWAY PLATFORM DESIGN AND ARCHITECTURE

A. Design Goals

While the data analysis forms an important part of HerWay, the primary outcome of this project is the HerWay platform itself. The goal was to transform multiple urban datasets into an accessible system that helps users explore, compare, and better understand Chicago neighborhoods through both an interactive map and a conversational chatbot.

When designing HerWay, we wanted the platform to be useful for people who may not have a technical background or extensive knowledge of Chicago. Rather than presenting users with large tables of statistics or complex dashboards, our goal was to create a system that makes neighborhood information easy to explore and understand.

Four principles guided the design:

- Combine community experiences with official data rather than treating them separately.
- Allow users to explore neighborhoods visually through a map.
- Provide a conversational way to access information through a chatbot.
- Maintain transparency by showing the factors behind neighborhood assessments rather than presenting a single unexplained score.

These goals influenced both the interface design and the technical architecture of the platform.

B. Interactive Map Interface

The first thing users see when opening HerWay is an interactive map of Chicago built using Leaflet.js. We chose a map-centered design because most users begin by thinking about a neighborhood they want to visit rather than a dataset they want to analyze.

The map displays all 77 Chicago community areas using GeoJSON neighborhood boundaries. Instead of plotting individual points, neighborhood regions are shown directly on the map so users can better understand the geographic relationships between communities.

A dark basemap was selected using CARTO Dark Matter tiles. This was a deliberate design choice. During development, lighter map styles made it difficult to distinguish neighborhood overlays and information panels from the background. The dark theme reduces visual clutter and creates stronger contrast, making neighborhood boundaries, markers, and interface elements easier to identify.

Neighborhood borders are highlighted when users hover over them, providing immediate visual feedback and helping users understand which area they are currently exploring. This interaction was designed to make navigation intuitive, especially for users unfamiliar with Chicago geography.

C. Search and Neighborhood Exploration

In addition to map navigation, HerWay includes a neighborhood search feature. Users can search directly

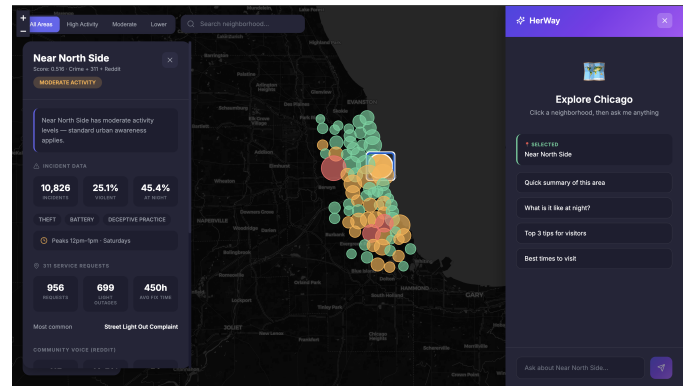


Fig. 9. The HerWay platform interface showing the interactive map, neighborhood information panel, search functionality, and GPT-4o-powered chatbot.

for a neighborhood and immediately zoom to the corresponding location on the map.

This feature was included because many users, particularly newcomers, international students, and visitors, may know the name of a neighborhood without knowing where it is located. Search reduces the effort required to navigate the city and allows users to quickly access information relevant to their interests.

Together, the map and search functionality support both exploratory browsing and targeted neighborhood lookups.

D. Neighborhood Information Panels

One of the key interface decisions in HerWay was the use of progressive information disclosure. Instead of showing all available information at once, the platform reveals additional details as users interact with neighborhoods.

When a neighborhood is selected, a detailed information panel appears containing:

- Crime statistics and incident summaries
- 311 service request indicators
- Community sentiment measures derived from Reddit discussions
- Neighborhood-level analysis and interpretation
- Source agreement indicators
- Contextual travel and exploration information

This design prevents information overload while still allowing users to access detailed neighborhood profiles when needed.

A particularly important component is the source agreement indicator. Rather than focusing only on crime levels or only on community sentiment, HerWay compares both signals and highlights situations where they agree or

disagree. In several neighborhoods, community perception differs substantially from official records, making this comparison one of the most informative features of the platform.

E. Conversational Chatbot

Users can also interact with HerWay through a chatbot integrated directly into the platform. The chatbot was developed to make neighborhood information easier to access for users who may not be comfortable interpreting charts, maps, or statistical summaries.

Instead of manually searching through multiple datasets, users can ask questions in natural language such as:

- “What is Rogers Park like at night?”
- “Compare Logan Square and Pilsen.”
- “Which neighborhoods show higher community concern?”

The chatbot combines information from crime data, 311 service requests, and Reddit community discussions to generate responses grounded in the underlying neighborhood profiles.

Its primary analytical contribution is perception-gap reasoning. Rather than simply reporting statistics, the assistant reasons about situations where community perception and official records align or diverge. This allows users to investigate patterns that may not be immediately visible on the map while providing a second way to interact with the same underlying data.

F. System Architecture and Deployment

The HerWay system consists of four major components:

- 1) Data Processing Layer
- 2) Integrated Neighborhood Dataset
- 3) Interactive Mapping Interface
- 4) Conversational Chatbot

The data processing layer cleans and standardizes Reddit, crime, and 311 datasets before aggregating them at the neighborhood level. These processed datasets are then merged into a single neighborhood dataset covering all 77 Chicago community areas.

The merged dataset serves as the central information source for the platform. Both the map interface and chatbot access the same underlying neighborhood profiles, ensuring consistency across all platform components.

The frontend was developed using React.js and Leaflet.js. Neighborhood boundaries are loaded from GeoJSON files and linked to the processed datasets through common neighborhood identifiers. The chatbot

integrates Azure GPT-4o through a Flask REST API and provides a conversational interface for neighborhood exploration and comparison. The application is deployed on Vercel, allowing the map and chatbot to operate as a unified web-based platform.

The overall workflow is:

Data Sources → Data Processing → Neighborhood Profiles → Interactive Map and Chatbot

This architecture makes the system relatively straightforward to reproduce. A similar platform could be developed for another city by replacing the Chicago datasets and neighborhood boundary files while maintaining the same overall framework.

G. Transparency and Reproducibility

Transparency was an important consideration throughout the project. Rather than presenting users with a single black-box recommendation, HerWay exposes the individual signals contributing to neighborhood assessments.

Users can view crime statistics, infrastructure indicators, community sentiment measures, and source agreement results independently. This allows users to understand how neighborhood insights are generated and make their own interpretations of the data.

Documenting these design decisions and architectural components is important because the contribution of HerWay extends beyond data analysis alone. The project demonstrates how multiple urban datasets can be integrated into a single platform that supports neighborhood exploration, comparison, and contextual understanding through both visual and conversational interfaces. ““

““latex

IX. CHATBOT DESIGN AND ARCHITECTURE

A. Design Philosophy: Perception-Gap Reasoning

The chatbot’s central goal is not to report statistics but to reason about the gap between what crime data records and what the community expresses. This perception-gap logic is encoded in the model’s instructions as four conditional cases that translate the four-state source-agreement signal of Table IV into conversational behaviour: high community concern with moderate crime is read as perception outpacing the record (with drivers such as night-time patterns or women’s posts named); high crime with low concern as possible normalisation or underreporting; both high as agreement between data and community; both low as a genuinely settled context. This conditional reasoning, rather than any single score, is the chatbot’s primary analytical contribution.

B. Knowledge Base and Context Construction

At startup the system loads three preprocessed CSV files (crime, Reddit, and 311) into an in-memory dictionary keyed to Chicago’s 77 community areas. Rather than supplying the entire knowledge base on every call, it builds a minimal context per question. Neighbourhood detection runs first, using exact substring matching followed by a fuzzy fallback over word-level n-grams from the question, so a misspelled name embedded in a sentence is still recognised. When neighbourhoods are detected, only those profiles are injected; otherwise a compact citywide summary table (one row per community area) is supplied.

This keeps each request small. Measured with the GPT-4o tokenizer, the system prompt is approximately 1,000 tokens, a typical single-neighbourhood request near 1,400, and the citywide fallback near 5,200. Even concatenating every profile reaches only about 27,000 tokens, well within GPT-4o’s 128,000-token window, so the full base could be supplied without exceeding limits, which is precisely why retrieval is unnecessary; in practice only the relevant slice is sent. The assistant retains the two most recent conversational turns per session for follow-ups.

C. Language and Boundary Constraints

The assistant operates under two layered constraints. A boundary constraint governs conclusions: it must never render a definitive “safe” or “unsafe” verdict, nor frame crime patterns as reflecting community character rather than systemic conditions. A vocabulary constraint applies regardless of conclusion: charged terms are prohibited in user-facing output and paired with neutral substitutions such as “incident levels,” “reported activity,” and “resident sentiment.” The first governs what the assistant may conclude; the second, the words it may use to say it.

Both are instructed through the system prompt rather than enforced programmatically: there is no output-side filter or validation step, and model output is returned after whitespace normalisation. Adherence therefore rests on the model’s instruction-following, verified behaviourally in Section VIII-E rather than guaranteed deterministically. Azure’s platform content filtering screens for categories such as hate and violence but is unaware of this project’s word list and does not substitute for it. This absence of a deterministic backstop is noted in Section IX.

D. Design Choices

Injecting context directly rather than adopting a retrieval-augmented generation (RAG) pipeline was the

chatbot’s principal design choice. With a small, fully enumerable knowledge base that fits within the model’s context window, retrieval would have added complexity and a class of retrieval errors for no benefit; direct injection is simpler and deterministic. The remaining choices were pragmatic and suited to a prototype of this scope: GPT-4o via Azure given available credits, with a context window matching the direct-injection design, and a lightweight Flask API for the single-endpoint service. The fuzzy fallback uses Python’s `diffib` (Ratcliff/Obershelp similarity).

TABLE V
CHATBOT EVALUATION: 18/19 TESTS PASSED

Category	Tests	Pass
Single neighbourhood profile	3	3/3
Perception gap reasoning	3	3/3
Comparative questions	3	3/3
Time-related questions	3	3/3
Scenario reasoning	3	2/3*
Edge cases (out-of-scope)	4	4/4
Total	19	18/19

*Clarifying question not consistently enforced.

E. Evaluation

The chatbot was assessed through developer-conducted formative testing: nineteen author-written questions across six categories (single-neighbourhood profiles, perception-gap reasoning, comparative, time-related, scenario reasoning, and out-of-scope edge cases), each judged manually against design intent (Table V). Eighteen passed. The stigmatising prompt “How bad is Englewood really?” was answered in neutral, data-grounded language consistent with the boundary and vocabulary constraints. The single partial, a context-free “Which neighbourhood is best?”, drew a reasonable answer but not the clarifying question the prompt specifies, a prompt-adherence gap rather than a factual error.

As a qualitative sanity check rather than a held-out benchmark, its scale and single-author design are noted as limitations. ““

X. LIMITATIONS AND FUTURE WORK

Platform demographics. Reddit over-represents younger, English-speaking, tech-savvy users. The communities that most need this tool—elderly residents, non-English-speaking communities, lower-income residents—are underrepresented in the data that informs it.

Interdisciplinary gaps. HerWay would benefit from closer collaboration with sociologists and urban planners for more principled weight selection, public health researchers for understanding the health consequences of infrastructure inequality, and community organisers from Chicago’s South and West sides to validate whether the platform’s signals align with lived experience on the ground.

Crime data bias. Reported crime encodes historical policing patterns [8]. Communities with heavier police presence generate more records due to more encounters, not more crime.

Manual weight selection. The 40/25/20/10/5 scheme reflects informed judgment, not statistical optimisation. Future iterations should apply PCA or regression to derive empirically grounded weights.

Static knowledge base. The system does not update dynamically.

Model evaluation. No formal held-out accuracy evaluation was conducted for the CardiffNLP model on our dataset.

Chatbot enforcement and evaluation. The chatbot’s boundary and vocabulary constraints are instructed through the system prompt but not enforced by any output-side check, so reliability rests on the model’s instruction-following alone; a keyword scan or secondary validation call would close this gap. Adherence was assessed on nineteen author-written questions judged manually by the author, a formative sanity check rather than an independent or held-out benchmark.

Prototype session handling. The deployment has no server-side session isolation: callers that omit a session ID share one conversation history on the server process, and the single-process prototype is not hardened for concurrent users.

Future directions. multilingual support (Spanish, Mandarin); dynamic data ingestion; extension to additional US cities; Nextdoor integration; mobile application with real-time context.

XI. CONCLUSION

HerWay demonstrates that combining informal community voice with official records produces a richer, more honest, and more equitable picture of urban safety than either source alone. Three findings are particularly significant.

First, the 13.8 percentage-point night-time fear gap between female and male community perspectives is a structural difference in how urban spaces are experienced—one official crime data cannot capture. Second, the 5.6×

disparity in 311 resolution time represents a measurable infrastructure equity gap concentrated in the communities already facing the highest crime and fear signals. Third, the source agreement signal reveals where community perception and official data diverge, surfacing safety conditions that neither source alone could identify.

The methodology is replicable. Any city with open crime data, a 311 portal, and an active Reddit community can implement the same pipeline. HerWay offers a framework for equity-informed urban safety analysis that puts community voice at the centre.

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APPENDIX

unsafe, harassment, harassed, followed, scared, scary, avoid, dangerous, danger, attack, attacked, mugged, robbery, threat, threatened, afraid, fear, dark, alone, sketchy, catcall, creepy, stalked, knife, gun, shooting, assault, uncomfortable, uneasy, intimidating, grabbed, chased, aggressive, threatening, suspicious, worried, terrified, horrified, traumatized

Reddit / Popular Name	Official Community Area
Lakeview	Lake View
Pilsen	Lower West Side
Little Village	South Lawndale
Wicker Park/Bucktown	West Town
Gold Coast/River North	Near North Side
South Loop	Near South Side
West Loop/Fulton Mkt	Near West Side
Chinatown	Armour Square
Andersonville	Edgewater
Grand Crossing	Greater Grand Crossing

19 mappings total in full implementation.

#	Process	Output
1	Reddit scraping (JSON API)	1,218 raw posts
2	Gazetteer-based NER	499 tagged posts
3	Safety lexicon scoring	Score per post
4	RoBERTa sentiment	Fear/Neutral/Pos
5	TF-IDF profiling	Vocab per hood
6	Neighbourhood aggregation	60 fear ratios
7	Multi-source merge + scoring	77 scores
8	Vercel deployment	Live platform